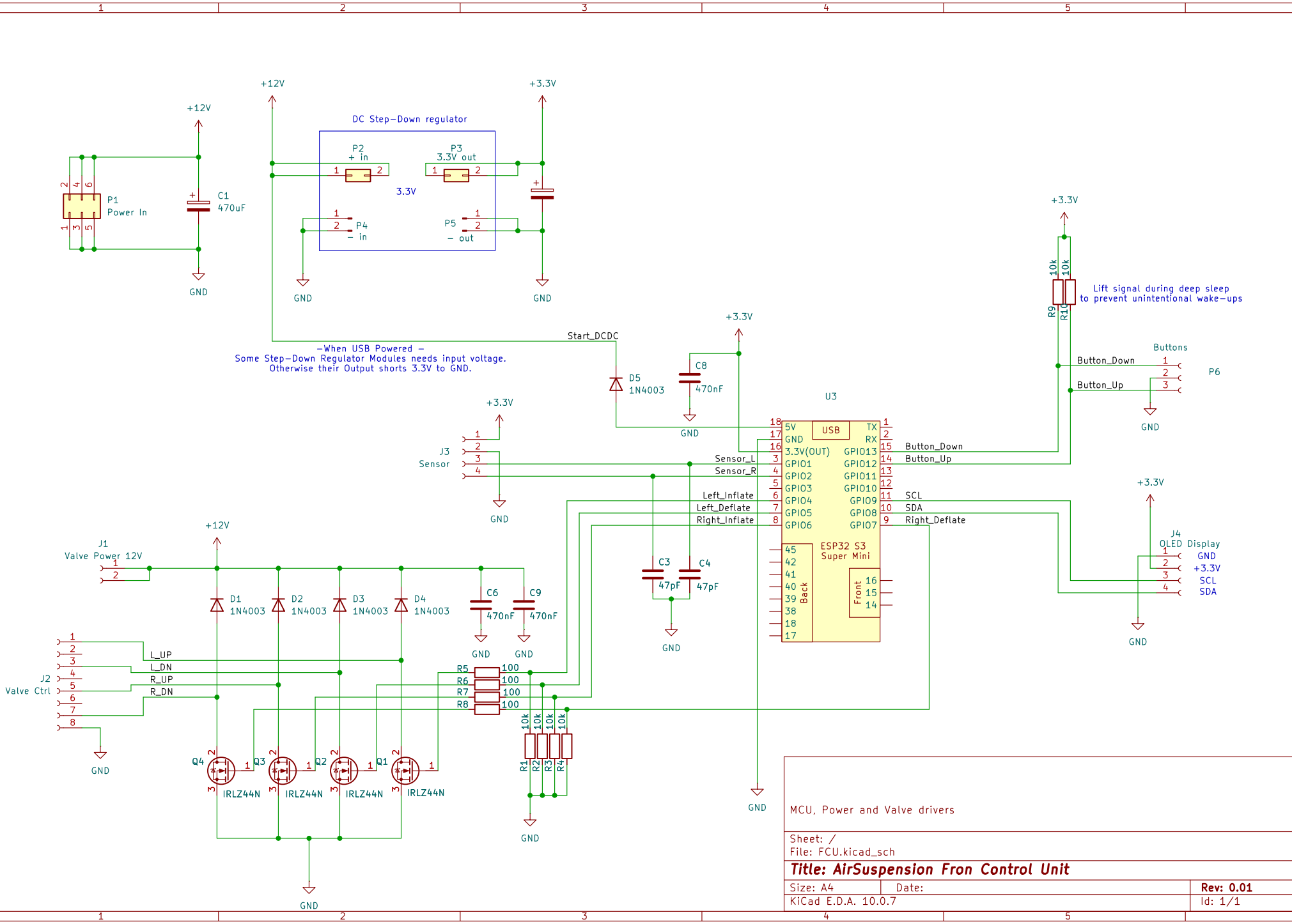

+12V
P1
Power In
C1
470uF
GND
DC Step-Down regulator
P2
+ in
P3
3.3V out
3.3V
P4
- in
P5
- out
+3.3V
When USB Powered -
Some Step-Down Regulator Modules needs input voltage.
Otherwise their Output shorts 3.3V to GND.
Start_DCDC
D5
1N4003
C8
470nF
U3
ESP32 S3
Super Mini
USB
5V
GND
3.3V(OUT)
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
TX
RX
GPIO13
GPIO12
GPIO11
GPIO10
GPIO9
GPIO8
GPIO7
Back
Front
Sensor_L
Sensor_R
Left_Inflate
Left_Deflate
Right_Inflate
Button_Down
Button_Up
SCL
SDA
Right_Deflate
J3
Sensor
C3
47pF
C4
47pF
R9
10k
R10
10k
Lift signal during deep sleep
to prevent unintentional wake-ups
Buttons
P6
J4
OLED Display
J1
Valve Power 12V
D1
1N4003
D2
1N4003
D3
1N4003
D4
1N4003
C6
470nF
C9
470nF
J2
Valve Ctrl
L_UP
L_DN
R_UP
R_DN
R5
100
R6
100
R7
100
R8
100
R1
10k
R2
10k
R3
10k
R4
10k
Q4
Q3
Q2
Q1
IRLZ44N
MCU, Power and Valve drivers
Sheet: /
File: FCU.kicad_sch
Title: AirSuspension Fron Control Unit
Size: A4
Date:
Rev: 0.01
KiCad E.D.A. 10.0.7
Id: 1/1